RESILIENCE REVIEW

Multi-Region Report: a second AWS region

What it takes to survive the loss of a whole AWS region for the accounts you choose: run cost, people by team, recovery time per service, cost and time to build, and where to put it.

1 service has no multi-region profile (Transform): costed by their catalog category, and their availability in other regions is not verified.

SCOPE
name contains 'prod'

ACCOUNTS GOING MULTI-REGION
40 of 40

PRIMARY REGION
Europe (London) (eu-west-2)

SECOND REGION
Europe (Frankfurt) (eu-central-1) · recommended

RECOVERY TARGET
Recover within 3 hours of a region-wide event

DATA RESIDENCY
Anywhere AWS has a region

1 Executive summary

BOTTOM LINE

To recover within 3 hours, a second region in Europe (Frankfurt) for 40 accounts would add roughly $2.5m to $6m a year to run (25 to 70% of their spend), cost $1.4m to $3.5m and take 5 to 10 months to build, and need 2 to 5 more people.

> **Figures are provisional.** They use default answers. Choose the recovery target and accounts, and answer the questions in section 2.

WHAT WE FOUND

- **Today.** 90% of regional spend runs in Europe (London); Europe (Ireland) already carries some.
- **Recommended region.** Europe (Frankfurt): 100% of spend is on services available there, about 15 ms from Europe (London), prices about +2% against it.
- **Runners-up.** Europe (Ireland): no Connect here ($20.9k/month to re-architect). US East (N. Virginia): ~80 ms further than Europe (Frankfurt).
- **Failover risk.** 5 of 40 chosen accounts run a material service that cannot meet the target as built (no live replica: restore takes hours).
- **Data residency.** AWS has no second region in the United Kingdom: any second region moves data across a border, which needs regulator and data-protection sign-off (UK GDPR; PRA SS2/21).
- **People.** About 2 to 5 extra FTE to run two regions ($400k to $900k a year): today's teams grow from 22 to about 26 people (+16%); Platform 10 to 11, Databases 3 to 3.8, DR 1 to 2.
- **Recovery time.** With pilot light, 8 services ($108.8k/month) would not be serving within 3 h: Redshift, Glue, EMR. See the recovery table.

WHAT WE RECOMMEND

1. **Settle residency first.** Agree with the regulator which data may sit outside the country.
2. **Tier the workloads.** Not every account needs the same target: keep instant failover for the few systems whose outage is a systemic event, and 3-hour recovery for the rest.
3. **Fix the hardest services first.** Start with services that have no live replica or are missing in the second region; they set the timeline.
4. **Rehearse.** Prove failover with a scheduled region-evacuation test and keep the evidence.

Extra cost to run (a year)

$2.5m to $6m PROVISIONAL

25 to 70% of the chosen accounts' spend

Cost to achieve (one-off)

$1.4m to $3.5m PROVISIONAL

4.5 to 10 person-years

Time to achieve

5 to 10 months PROVISIONAL

bound by team capacity

Extra people

2 to 5 FTE PROVISIONAL

$400k to $900k a year

Recommended region

Europe (Frankfurt)

~15 ms · 100% service parity

Accounts with failover risk

5 of 40

a material service cannot meet the target as built

2 Your choices

Saved in this browser only. These answers belong to this report alone; the exit risk report has its own.

PROVISIONAL Using default answers. Change any answer, or confirm these.

Recovery target after a region-wide event

Recover within 3 hours

Systemic risk event: data replicated continuously, a minimal footprint scaled up on failover (pilot light).

Where may the second region be?

Anywhere AWS has a region

Primary region

Detected: Europe (London)

Second region

Recommended: Europe (Frankfurt)

AWS Support plan

Enterprise

Application engineers available (FTE)

10

% of platform and data time on the build

50

Blended day rate

750

Currency

USD

Cost per extra FTE a year (0 = day rate × 220)

0

Standard CI/CD pipeline?

Several, partially standard pipelines

Terraform, and is it up to date?

Terraform with significant drift

Regulated (adds evidence and sign-off)

Your teams today people now, so the report can show what a second region adds to each

Platform, cloud and SRE

10

Databases and data

3

Cloud security and IAM

3

Application support and ops

5

DR and resilience

1

3 **Accounts to make multi-region**

Tick the accounts that need a second region, or pick an OU. Sorted by spend. Readiness is the worst failover risk among an account's material services for the chosen recovery target; open an account to see each service's risk.

Account	Spend / month	Readiness	Multi-region risks
data-platform-prod 561017993421 · Root / Workloads / Production	$133k	High	▾ High 5 · Moderate 3 · Low 2 **High** **Redshift** $62k/mo · No live replica: restore or re-run No live cross-region replica. Copy snapshots to the second region and restore (hours for a large warehouse), or load both regions from the pipelines. **High** **Glue** $15k/mo · No live replica: restore or re-run Jobs, crawlers and the Data Catalog are regional: replicate the catalog and re-run pipelines in the second region. **High** **EMR** $11k/mo · No live replica: restore or re-run Clusters are rebuilt in the second region from code; data must be replicated to S3 there. **High** **Athena** $6.6k/mo · No live replica: restore or re-run Queries need the data and Glue catalog to exist in the second region. **High** **MWAA (Airflow)** $2.0k/mo · No live replica: restore or re-run Data platforms are built single-region. Datasets, catalogs and jobs must be copied and pipelines re-run; recovery takes hours. **Moderate** **QuickSight** $5.1k/mo · Manual or scripted failover Has its own failover model, often only for specific region pairs. **Moderate** **EC2 Other (EBS, NAT, transfer)** $3.7k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Moderate** **Firehose** $2.6k/mo · Manual or scripted failover Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over. **Low** **S3** $22k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. **Low** **Lake Formation** $640/mo · AWS replicates and fails over Pipelines must deploy to both regions; the tooling itself can stay in one.
core-banking-prod 813958344723 · Root / Workloads / Production	$122k	Moderate	▾ Moderate 7 · Low 1 **Moderate** **EC2** $56k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $34k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EC2 Other (EBS, NAT, transfer)** $15k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Moderate** **FSx** $5.7k/mo · Manual or scripted failover Cross-region backup copy, or SnapMirror for FSx for NetApp ONTAP; restore or break the mirror to fail over. **Moderate** **Direct Connect** $4.6k/mo · Manual or scripted failover Needs connections or a Direct Connect gateway association into the second region; physical lead times are weeks to months. **Moderate** **ELB** $2.0k/mo · Manual or scripted failover Load balancers are regional; Route 53 health checks switch traffic between them. **Moderate** **Amazon MQ** $1.4k/mo · Manual or scripted failover Needs a cross-region replica. Asynchronous replication loses the last seconds of writes, and promoting the replica is a decision someone has to make. **Low** **Backup** $3.4k/mo · AWS replicates and fails over Copy backups to a vault in the second region; restore time grows with data size.

Account	Spend / month	Readiness	Multi-region risks
payments-core-prod 534439589175 · Root / Workloads / Production	$116k	Moderate	▾ Moderate 10 · Low 2 **Moderate** **EC2** $42k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $28k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EC2 Other (EBS, NAT, transfer)** $9.9k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Moderate** **MSK (Kafka)** $7.9k/mo · Manual or scripted failover MSK Replicator copies topics cross-region; consumers must handle offset translation on failover. **Moderate** **EKS** $4.6k/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Moderate** **ElastiCache** $4.2k/mo · Manual or scripted failover Global Datastore (Redis OSS / Valkey) replicates cross-region; without it the cache starts cold after failover. **Moderate** **CloudWatch** $3.6k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region. **Moderate** **ELB** $2.3k/mo · Manual or scripted failover Load balancers are regional; Route 53 health checks switch traffic between them. **Moderate** **Lambda** $1.2k/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. **Moderate** **KMS** $978/mo · Manual or scripted failover Keys are regional. Data encrypted with single-Region keys cannot be decrypted in the second region: use multi-Region keys and re-encrypt. **Low** **S3** $4.5k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. **Low** **DynamoDB** $2.8k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer.

Account	Spend / month	Readiness	Multi-region risks
cards-processing-prod 966773840736 · Root / Workloads / Production	$101k	High	▾ High 1 · Moderate 8 · Low 2 **High** **Cognito** $2.4k/mo · No live replica: restore or re-run User pools are regional with no replication and password hashes cannot be exported: users must be migrated or re-registered, or a sync built. **Moderate** **Lambda** $23k/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. **Moderate** **API Gateway** $9.8k/mo · Manual or scripted failover Deploy the API in both regions behind a custom domain with Route 53 failover. **Moderate** **Kinesis Streams** $8.4k/mo · Manual or scripted failover No native cross-region replication: producers dual-write or a consumer replicates; consumers restart at a new position. **Moderate** **Step Functions** $7.1k/mo · Manual or scripted failover State machines are regional; running executions do not fail over and must be restarted. **Moderate** **CloudWatch** $4.8k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region. **Moderate** **SQS** $3.2k/mo · Manual or scripted failover Queues are regional and messages are not replicated: in-flight messages are lost or delayed on failover. **Moderate** **EventBridge** $1.7k/mo · Manual or scripted failover Global endpoints can fail event buses over between two regions; rules and targets must exist in both. **Moderate** **KMS** $1.4k/mo · Manual or scripted failover Keys are regional. Data encrypted with single-Region keys cannot be decrypted in the second region: use multi-Region keys and re-encrypt. **Low** **DynamoDB** $35k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer. **Low** **S3** $4.3k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
digital-channels-prod 978681853473 · Root / Workloads / Production	$57k	High	▾ High 1 · Moderate 8 · Low 1 **High** **Cognito** $5.6k/mo · No live replica: restore or re-run User pools are regional with no replication and password hashes cannot be exported: users must be migrated or re-registered, or a sync built. **Moderate** **ECS** $10k/mo · Manual or scripted failover Clusters and services are regional: deploy to both regions and fail over with Route 53. **Moderate** **RDS / Aurora** $9.6k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EC2** $8.0k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **AppSync** $4.3k/mo · Manual or scripted failover Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over. **Moderate** **Pinpoint** $2.8k/mo · Manual or scripted failover Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over. **Moderate** **SNS** $1.8k/mo · Manual or scripted failover Topics and subscriptions are regional: recreate in the second region and publish there. **Moderate** **WAF** $1.5k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region. **Moderate** **Amplify** $1.3k/mo · Manual or scripted failover Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over. **Low** **S3** $2.5k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
ml-platform-prod 217230843944 · Root / Workloads / Production	$47k	Moderate	▾ Moderate 6 · Low 2 **Moderate** **SageMaker** $19k/mo · Manual or scripted failover Models and endpoints must be redeployed in the second region; training data replicated. **Moderate** **EC2** $6.6k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **OpenSearch** $4.7k/mo · Manual or scripted failover Cross-cluster replication to a domain in the second region; re-point clients on failover. **Moderate** **Textract** $3.0k/mo · Manual or scripted failover Models, endpoints and quotas differ by region and must be provisioned in the second region. **Moderate** **Bedrock AgentCore** $1.2k/mo · Manual or scripted failover Models, endpoints and quotas differ by region and must be provisioned in the second region. **Moderate** **Transform** $495/mo · Manual or scripted failover No cross-region failover built in. Images, launch templates and capacity must exist in the second region and be started by a runbook or automation. **Low** **Bedrock** $7.6k/mo · AWS replicates and fails over Cross-region inference spreads requests across regions, but model availability differs: check every model used exists in the second region. **Low** **S3** $4.2k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
shared-network-prod 263737281768 · Root / Infrastructure	$37k	Moderate	▾ Moderate 4 **Moderate** **VPC** $12k/mo · Manual or scripted failover VPCs, subnets, endpoints and inter-region peering or Transit Gateway must be built in the second region. **Moderate** **EC2 Other (EBS, NAT, transfer)** $8.3k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Moderate** **Direct Connect** $6.6k/mo · Manual or scripted failover Needs connections or a Direct Connect gateway association into the second region; physical lead times are weeks to months. **Moderate** **Network Firewall** $5.3k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region.
contact-centre-prod 125285552493 · Root / Workloads / Production	$34k	Moderate	▾ Moderate 5 · Low 2 **Moderate** **Connect** $21k/mo · Manual or scripted failover Connect Global Resiliency replicates instances and numbers between specific region pairs and is enabled by AWS on request; otherwise a second instance with its own numbers. **Moderate** **Transcribe** $3.4k/mo · Manual or scripted failover Models, endpoints and quotas differ by region and must be provisioned in the second region. **Moderate** **Lex** $3.1k/mo · Manual or scripted failover Models, endpoints and quotas differ by region and must be provisioned in the second region. **Moderate** **Lambda** $2.3k/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. **Moderate** **Kinesis Streams** $1.3k/mo · Manual or scripted failover No native cross-region replication: producers dual-write or a consumer replicates; consumers restart at a new position. **Low** **DynamoDB** $1.8k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer. **Low** **S3** $1.5k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
fraud-analytics-prod 466806226017 · Root / Workloads / Production	$20k	High	▾ High 1 · Moderate 4 · Low 1 **High** **Managed Flink** $3.6k/mo · No live replica: restore or re-run Data platforms are built single-region. Datasets, catalogs and jobs must be copied and pipelines re-run; recovery takes hours. **Moderate** **Neptune** $6.8k/mo · Manual or scripted failover Global Database for cross-region replicas; failover is a promotion step. **Moderate** **SageMaker** $4.3k/mo · Manual or scripted failover Models and endpoints must be redeployed in the second region; training data replicated. **Moderate** **Timestream** $2.0k/mo · Manual or scripted failover Needs a cross-region replica. Asynchronous replication loses the last seconds of writes, and promoting the replica is a decision someone has to make. **Moderate** **Lambda** $994/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. **Low** **DynamoDB** $2.6k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer.
doc-store-prod 454519794043 · Root / Workloads / Production	$9.7k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $4.8k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $2.9k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $986/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $975/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
id-verify-prod 267387487076 · Root / Workloads / Production	$8.7k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $4.4k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $2.6k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $877/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $877/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
investments-prod 514740085648 · Root / Workloads / Production	$8.5k	Moderate	▾ Moderate 2 · Low 1 **Moderate** **EC2** $5.1k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **EC2 Other (EBS, NAT, transfer)** $1.7k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Low** **S3** $1.7k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
mobile-backend-prod 434423300025 · Root / Workloads / Production	$8.0k	Moderate	▾ Moderate 4 Moderate **RDS / Aurora** $2.8k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EC2** $2.4k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **ECS** $1.6k/mo · Manual or scripted failover Clusters and services are regional: deploy to both regions and fail over with Route 53. Moderate **Lambda** $1.2k/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.
identity-prod 189183446249 · Root / Infrastructure	$7.9k	Moderate	▾ Moderate 3 Low 1 Moderate **EC2** $3.4k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **Directory Service** $2.2k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region. Moderate **ACM / Private CA** $1.6k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region. Low **Secrets Manager** $647/mo · AWS replicates and fails over Turn on secret replication to the second region.
payroll-api-prod 295250496497 · Root / Workloads / Production	$7.6k	Moderate	▾ Moderate 3 Low 1 Moderate **EC2** $3.8k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $2.3k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $758/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $755/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
branch-queue-prod 717640922470 · Root / Workloads / Production	$7.3k	Moderate	▾ Moderate 3 Low 1 Moderate **EC2** $3.7k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $2.2k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $732/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $728/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
insurance-quotes-prod 498633168856 · Root / Workloads / Production	$7.2k	Moderate	▾ Moderate 3 Low 1 Moderate **EC2** $3.6k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $2.2k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $715/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $719/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
reporting-hub-prod 244219422241 · Root / Workloads / Production	$7.1k	Moderate	▾ Moderate 2 · Low 1 Moderate **EC2** $4.3k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **EC2 Other (EBS, NAT, transfer)** $1.4k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. Low **S3** $1.4k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
merchant-portal-prod 386906790760 · Root / Workloads / Production	$7.0k	Moderate	▾ Moderate 4 Moderate **RDS / Aurora** $2.5k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EC2** $2.1k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **ECS** $1.4k/mo · Manual or scripted failover Clusters and services are regional: deploy to both regions and fail over with Route 53. Moderate **Lambda** $1.1k/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.
disputes-prod 844781801768 · Root / Workloads / Production	$6.9k	Moderate	▾ Moderate 3 · Low 1 Moderate **EC2** $3.5k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $2.1k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $685/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $681/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
card-controls-prod 170174047247 · Root / Workloads / Production	$6.4k	Moderate	▾ Moderate 3 · Low 1 Moderate **EC2** $3.2k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $1.9k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $642/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $638/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
pricing-prod 322790187289 · Root / Workloads / Production	$6.1k	Moderate	▾ Moderate 3 · Low 1 Moderate **EC2** $3.0k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $1.8k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $613/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $609/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
collections-prod 110356959844 · Root / Workloads / Production	$6.0k	Moderate	▾ Moderate 2 · Low 1 Moderate **EC2** $3.6k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **EC2 Other (EBS, NAT, transfer)** $1.2k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. Low **S3** $1.2k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
audit-trail-prod 252831874844 · Root / Workloads / Production	$5.4k	Moderate	▾ Moderate 3 · Low 1 Moderate **EC2** $2.7k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $1.6k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $533/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $537/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
loans-origination-prod 369887811678 · Root / Workloads / Production	$5.3k	Moderate	▾ Moderate 3 · Low 1 Moderate **EC2** $2.7k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $1.6k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $529/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $530/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
sms-gateway-prod 698441060454 · Root / Workloads / Production	$4.7k	Moderate	▾ Moderate 3 · Low 1 Moderate **EC2** $2.4k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $1.4k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $470/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $471/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
kyc-checks-prod 706271427442 · Root / Workloads / Production	$4.5k	Moderate	▾ Moderate 3 · Low 1 Moderate **EC2** $2.3k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $1.3k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $450/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $453/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
atm-switch-prod 268805426379 · Root / Workloads / Production	$4.5k	Moderate	▾ Moderate 2 · Low 1 **Moderate** **EC2** $2.7k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **EC2 Other (EBS, NAT, transfer)** $896/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Low** **S3** $893/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
legacy-batch-prod 107537456074 · Root / Workloads / Production	$4.2k	High	▾ High 1 · Moderate 2 · Low 1 **High** **Data Pipeline** $327/mo · No live replica: restore or re-run Data platforms are built single-region. Datasets, catalogs and jobs must be copied and pipelines re-run; recovery takes hours. **Moderate** **EC2** $2.6k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **SWF** $712/mo · Manual or scripted failover Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over. **Low** **S3** $536/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
search-prod 899894212661 · Root / Workloads / Production	$4.0k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **Lambda** $1.4k/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. **Moderate** **API Gateway** $793/mo · Manual or scripted failover Deploy the API in both regions behind a custom domain with Route 53 failover. **Moderate** **SQS** $396/mo · Manual or scripted failover Queues are regional and messages are not replicated: in-flight messages are lost or delayed on failover. **Low** **DynamoDB** $1.4k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer.
geo-services-prod 154224630493 · Root / Workloads / Production	$4.0k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $2.0k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $1.2k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $401/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $397/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
chatbot-prod 925052687979 · Root / Workloads / Production	$3.4k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $1.7k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $1.0k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $343/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $342/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

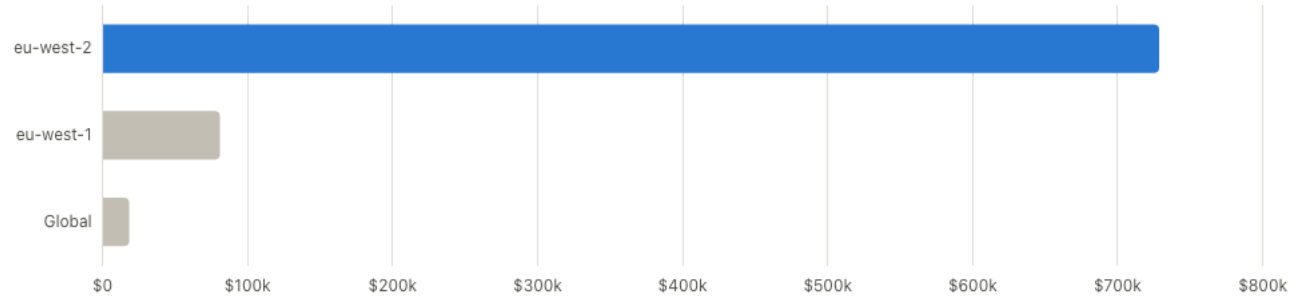
Account	Spend / month	Readiness	Multi-region risks
statements-prod 672004302296 · Root / Workloads / Production	$2.9k	Moderate	▾ Moderate 4 Moderate **RDS / Aurora** $1.0k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EC2** $881/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **ECS** $591/mo · Manual or scripted failover Clusters and services are regional: deploy to both regions and fail over with Route 53. Moderate **Lambda** $438/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.
treasury-reporting-prod 499647302627 · Root / Workloads / Production	$2.8k	Moderate	▾ Moderate 3 Low 2 Moderate **QuickSight** $781/mo · Manual or scripted failover Has its own failover model, often only for specific region pairs. Moderate **Lambda** $533/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. Moderate **Step Functions** $287/mo · Manual or scripted failover State machines are regional; running executions do not fail over and must be restarted. Low **DynamoDB** $1.0k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer. Low **S3** $223/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
notifications-prod 117309034122 · Root / Workloads / Production	$2.6k	Moderate	▾ Moderate 3 Low 1 Moderate **EC2** $1.3k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $782/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $260/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $263/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
partner-api-prod 283827598105 · Root / Workloads / Production	$2.2k	Moderate	▾ Moderate 3 Low 1 Moderate **EC2** $1.1k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $664/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $222/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $223/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
onboarding-prod 149964263026 · Root / Workloads / Production	$2.1k	Moderate	▾ Moderate 4 Moderate **RDS / Aurora** $722/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EC2** $627/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **ECS** $414/mo · Manual or scripted failover Clusters and services are regional: deploy to both regions and fail over with Route 53. Moderate **Lambda** $315/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.
rewards-prod 271987579183 · Root / Workloads / Production	$1.7k	Moderate	▾ Moderate 3 Low 1 Moderate **EC2** $829/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $497/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $166/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $164/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
limits-prod 152477260104 · Root / Workloads / Production	$1.4k	Moderate	▾ Moderate 2 Low 1 Moderate **EC2** $867/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **EC2 Other (EBS, NAT, transfer)** $292/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. Low **S3** $289/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
fx-rates-prod 753954983820 · Root / Workloads / Production	$1.0k	Moderate	▾ Moderate 3 Low 1 Moderate **EC2** $518/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **RDS / Aurora** $313/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EKS** $105/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. Low **S3** $104/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

4 Which second region

Where the chosen accounts run today

Monthly spend by region.



Recommended: Europe (Frankfurt)

Best balance of service parity, latency and price for the chosen accounts and recovery target.

Region	Europe (Frankfurt) (eu-central-1)
Estimated latency from primary	~15 ms round trip
Service parity (by spend)	100.0%
Not available there	None
Prices against primary	+2%
Already runs there	0.0% of regional spend

Candidate regions ranked by service parity, latency and price

Region	Latency	Parity	Not available	Price	Score	Why it ranks here
Europe (Frankfurt) eu-central-1 **Recommended**	~15 ms	100.0%	none	+2%	87	Best balance of service parity, latency and price.
Europe (Ireland) eu-west-1	~10 ms	97.4%	Connect	-4%	86	No Connect here ($20.9k/month to re-architect). 10% of spend already runs here.
US East (N. Virginia) us-east-1	~95 ms	100.0%	none	-14%	84	~80 ms further than Europe (Frankfurt).
Canada (Central) ca-central-1	~85 ms	100.0%	none	-5%	83	~70 ms further than Europe (Frankfurt).
US West (Oregon) us-west-2	~120 ms	100.0%	none	-14%	82	~105 ms further than Europe (Frankfurt).
Europe (Stockholm) eu-north-1	~25 ms	96.7%	Connect, Lex, Textract	-9%	79	No Connect, Lex, Textract here ($26.9k/month to re-architect); ~10 ms further than Europe (Frankfurt).
Europe (Paris) eu-west-3	~10 ms	97.0%	Connect, Lex	+1%	78	No Connect, Lex here ($24.0k/month to re-architect).
Europe (Spain) eu-south-2	~20 ms	96.4%	Amplify, Connect, Lex, Transcribe	-3%	77	No Amplify, Connect, Lex and others here ($28.7k/month to re-architect).
Asia Pacific (Seoul) ap-northeast-2	~140 ms	100.0%	none	+5%	76	~125 ms further than Europe (Frankfurt); about 3% dearer than Europe (Frankfurt).
US East (Ohio) us-east-2	~100 ms	97.0%	Connect, Lex	-14%	75	No Connect, Lex here ($24.0k/month to re-architect); ~85 ms further than Europe (Frankfurt).

Regulation to check before data moves to Europe (Frankfurt); pointers, not legal advice

UK GDPR (international transfers)

Outside the UK, a transfer to Germany needs UK adequacy regulations or safeguards (the International Data Transfer Agreement or the UK Addendum to the EU clauses).

PRA SS2/21 (outsourcing and third-party risk management)

Notify the PRA of material outsourcing changes, which include where data is held, and include the second region in exit and stressed-exit plans.

5 Cost to run, and the people

Second-region infrastructure

$2m to $4.5m PROVISIONAL

a year, incl. replication traffic

AWS Support uplift

$60k to $140k PROVISIONAL

Enterprise plan, a year

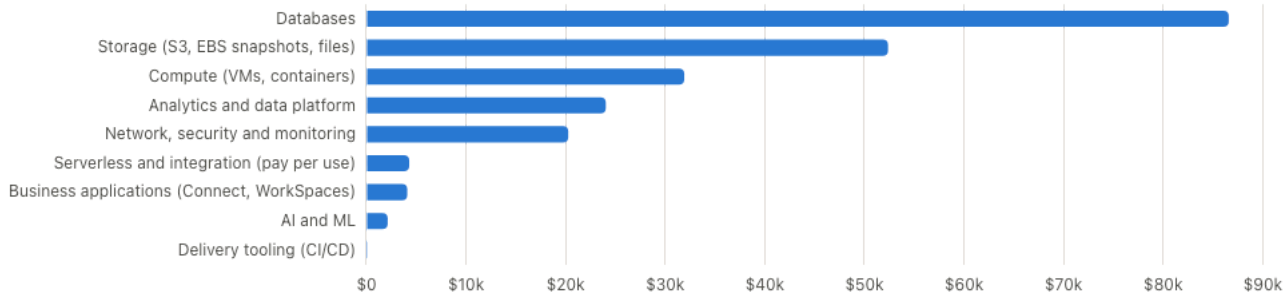
Extra people

$400k to $900k PROVISIONAL

2 to 5 FTE a year

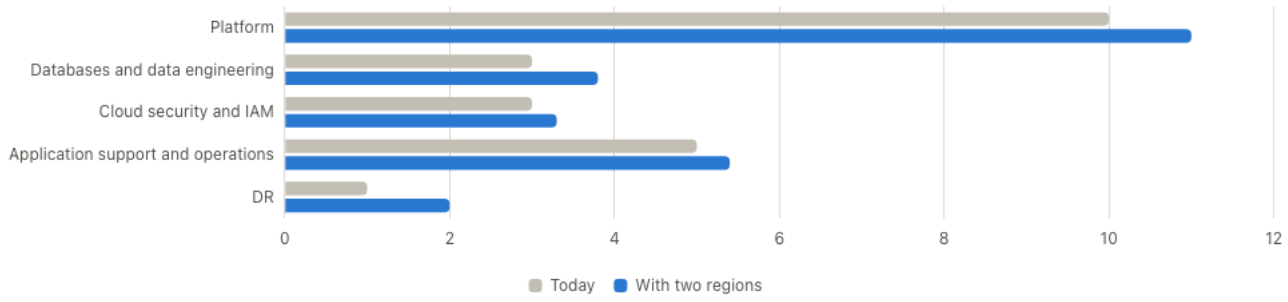
Second-region infrastructure by service class

Monthly, after the recovery target's share and regional prices.



Team sizes: today and with two regions

People, central estimate.



What a second region does to each team

Team	Today	Extra	With two regions	Change
Platform, cloud engineering and SRE On-call, patching and changes in two regions; two of every shared component.	10	+1.0	11.0	+10%
Databases and data engineering Replication, consistency checks, failover and failback of the data.	3	+0.8	3.8	+28%
Cloud security and IAM Guardrails, keys, logging and monitoring in the second region.	3	+0.3	3.3	+10%
Application support and operations Runbooks, deployments and support for each account in two regions.	5	+0.4	5.4	+8%
DR, continuity and resilience Failover tests, game days and the evidence regulators ask for.	1	+1.0	2.0	+100%
All teams (extra quoted as 2 to 5 FTE)	**22**	**+3.5**	**25.5**	**+16%**

Central estimate, at $165k per extra FTE a year. Set today's team sizes in section 2.

6 Recovery time by service

Your recovery target

within 3 h

Pilot light, from section 2

Services that meet it

47 of 55

as built today, with that strategy

Services that miss it

8

$109k/month of spend

Estimated time to be serving again (RTO) and data that could be lost (RPO) in the second region, for each service in the chosen accounts and each DR strategy. Your target's strategy is highlighted. Indicative: prove each one with a failover test.

Service	Spend / month	Failover	RTO: Backup and restore	RTO: Pilot light	RTO: Warm standby	RTO: Active-active	RPO: Pilot light	Meets 3
Redshift 1 account	$62k	No live replica: restore or re-run	2 to 7 days	**10 h to 2 days**	5 to 20 h	2 to 10 h	**1 to 24 h**	No needs a redesig
Glue 1 account	$15k	No live replica: restore or re-run	30 h to 4 days	**5 to 30 h**	2 to 10 h	1.2 to 5 h	**1 to 24 h**	No needs a redesig
EMR 1 account	$11k	No live replica: restore or re-run	25 h to 3 days	**4 to 25 h**	2 to 8 h	1 to 4 h	**1 to 24 h**	No needs a redesig
Cognito 2 accounts	$8.0k	No live replica: restore or re-run	24 h to 3 days	**4 to 24 h**	2 to 8 h	1 to 4 h	**1 to 24 h**	No needs a redesig
Athena 1 account	$6.6k	No live replica: restore or re-run	24 h to 3 days	**4 to 24 h**	2 to 8 h	1 to 4 h	**1 to 24 h**	No needs a redesig
Managed Flink 1 account	$3.6k	No live replica: restore or re-run	24 h to 3 days	**4 to 24 h**	2 to 8 h	1 to 4 h	**1 to 24 h**	No needs a redesig
MWAA (Airflow) 1 account	$2.0k	No live replica: restore or re-run	24 h to 3 days	**4 to 24 h**	2 to 8 h	1 to 4 h	**1 to 24 h**	No needs a redesig
Data Pipeline 1 account	$327	No live replica: restore or re-run	24 h to 3 days	**4 to 24 h**	2 to 8 h	1 to 4 h	**1 to 24 h**	No needs a redesig
EC2 33 accounts	$188k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
RDS / Aurora 25 accounts	$107k	Manual or scripted failover	24 h to 3 days	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
S3 31 accounts	$55k	AWS replicates and fails over	5 to 19 h	**15 min to 1 h**	5 to 30 min	up to 5 min	**up to 5 min**	Yes
DynamoDB 6 accounts	$44k	AWS replicates and fails over	4 to 17 h	**15 min to 1 h**	5 to 30 min	up to 5 min	**up to 5 min**	Yes
EC2 Other (EBS, NAT, transfer) 9 accounts	$43k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Lambda 10 accounts	$33k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
SageMaker 2 accounts	$23k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Connect 1 account	$21k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes

Service	Spend / month	Failover	RTO: Backup and restore	RTO: Pilot light	RTO: Warm standby	RTO: Active-active	RPO: Pilot light	Meets 3
ECS 5 accounts	$14k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
EKS 19 accounts	$14k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
VPC 1 account	$12k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Direct Connect 2 accounts	$11k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
API Gateway 2 accounts	$11k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Kinesis Streams 2 accounts	$9.7k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
CloudWatch 2 accounts	$8.4k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
MSK (Kafka) 1 account	$7.9k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Bedrock 1 account	$7.6k	AWS replicates and fails over	2 to 8 h	**15 min to 1 h**	5 to 30 min	up to 5 min	**up to 5 min**	Yes
Step Functions 2 accounts	$7.4k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Neptune 1 account	$6.8k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
QuickSight 2 accounts	$5.9k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
FSx 1 account	$5.7k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Network Firewall 1 account	$5.3k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
OpenSearch 1 account	$4.7k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
AppSync 1 account	$4.3k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes

Service	Spend / month	Failover	RTO: Backup and restore	RTO: Pilot light	RTO: Warm standby	RTO: Active-active	RPO: Pilot light	Meets 3
ELB 2 accounts	$4.2k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
ElastiCache 1 account	$4.2k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
SQS 2 accounts	$3.6k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Transcribe 1 account	$3.4k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Backup 1 account	$3.4k	AWS replicates and fails over	2 to 8 h	**15 min to 1 h**	5 to 30 min	up to 5 min	**up to 5 min**	Yes
Lex 1 account	$3.1k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Textract 1 account	$3.0k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Pinpoint 1 account	$2.8k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Firehose 1 account	$2.6k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
KMS 2 accounts	$2.4k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Directory Service 1 account	$2.2k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
Timestream 1 account	$2.0k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
SNS 1 account	$1.8k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
EventBridge 1 account	$1.7k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
ACM / Private CA 1 account	$1.6k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes
WAF 1 account	$1.5k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	Yes

Service	Spend / month	Failover	RTO: Backup and restore	RTO: Pilot light	RTO: Warm standby	RTO: Active-active	RPO: Pilot light	Meets 3
Amazon MQ 1 account	$1.4k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	**Yes**
Amplify 1 account	$1.3k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	**Yes**
Bedrock AgentCore 1 account	$1.2k	Manual or scripted failover	8 to 24 h	**30 min to 3 h**	30 min to 2 h	up to 15 min	**5 min to 1 h**	**Yes**

Services under $1.0k/month that meet the target are left out (4).

7 Cost and time to achieve

Cost to achieve (one-off)

$1.4m to $3.5m PROVISIONAL

engineering, plus running the region while it is built

Time to achieve

5 to 10 months PROVISIONAL

bound by team capacity · team of 16.5 full-time people

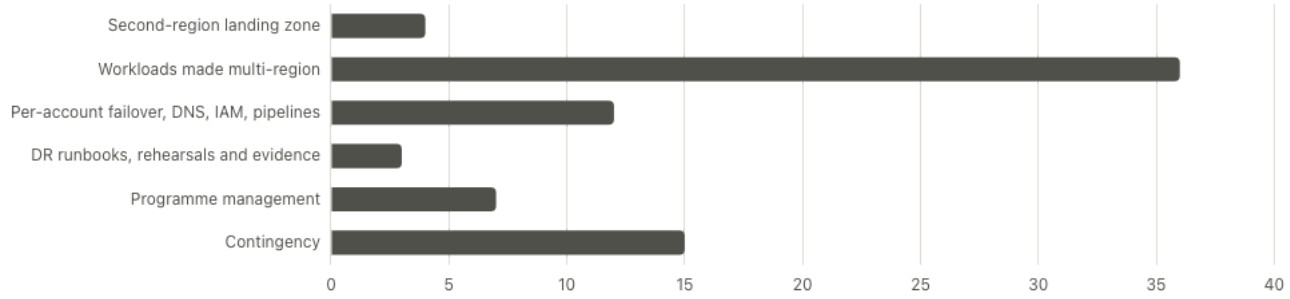
Engineering effort

4.5 to 10 person-years PROVISIONAL

40 material accounts

Where the effort goes

Person-months, after the delivery-maturity adjustment.



8 Service parity in the second region

Service	Class	Accounts	Spend / month	In eu-central-1
EC2	Compute (VMs, containers)	33	$188k	Available
RDS / Aurora	Databases	25	$107k	Available
Redshift	Analytics and data platform	1	$62k	Available
S3	Storage (S3, EBS snapshots, files)	31	$55k	Available
DynamoDB	Databases	6	$44k	Available
EC2 Other (EBS, NAT, transfer)	Compute (VMs, containers)	9	$43k	Available
Lambda	Serverless and integration (pay per use)	10	$33k	Available
SageMaker	AI and ML	2	$23k	Available
Connect	Business applications (Connect, WorkSpaces)	1	$21k	Available
Glue	Analytics and data platform	1	$15k	Available
ECS	Compute (VMs, containers)	5	$14k	Available
EKS	Compute (VMs, containers)	19	$14k	Available
VPC	Network, security and monitoring	1	$12k	Available
Direct Connect	Network, security and monitoring	2	$11k	Available
API Gateway	Serverless and integration (pay per use)	2	$11k	Available
EMR	Analytics and data platform	1	$11k	Available
Kinesis Streams	Databases	2	$9.7k	Available
CloudWatch	Network, security and monitoring	2	$8.4k	Available
Cognito	Serverless and integration (pay per use)	2	$8.0k	Available
MSK (Kafka)	Databases	1	$7.9k	Available
Bedrock	AI and ML	1	$7.6k	Available
Step Functions	Serverless and integration (pay per use)	2	$7.4k	Available
Neptune	Databases	1	$6.8k	Available
Athena	Serverless and integration (pay per use)	1	$6.6k	Available
QuickSight	Business applications (Connect, WorkSpaces)	2	$5.9k	Available
FSx	Storage (S3, EBS snapshots, files)	1	$5.7k	Available
Network Firewall	Network, security and monitoring	1	$5.3k	Available
OpenSearch	Databases	1	$4.7k	Available
AppSync	Serverless and integration (pay per use)	1	$4.3k	Available
ELB	Network, security and monitoring	2	$4.2k	Available
ElastiCache	Databases	1	$4.2k	Available
Managed Flink	Analytics and data platform	1	$3.6k	Available
SQS	Serverless and integration (pay per use)	2	$3.6k	Available
Transcribe	AI and ML	1	$3.4k	Available
Backup	Storage (S3, EBS snapshots, files)	1	$3.4k	Available
Lex	AI and ML	1	$3.1k	Available
Textract	AI and ML	1	$3.0k	Available
Pinpoint	Serverless and integration (pay per use)	1	$2.8k	Available
Firehose	Serverless and integration (pay per use)	1	$2.6k	Available

Service	Class	Accounts	Spend / month	In eu-central-1
KMS	Network, security and monitoring	2	$2.4k	Available
Directory Service	Network, security and monitoring	1	$2.2k	Available
MWAA (Airflow)	Analytics and data platform	1	$2.0k	Available
Timestream	Databases	1	$2.0k	Not verified
SNS	Serverless and integration (pay per use)	1	$1.8k	Available
EventBridge	Serverless and integration (pay per use)	1	$1.7k	Available
ACM / Private CA	Network, security and monitoring	1	$1.6k	Available
WAF	Network, security and monitoring	1	$1.5k	Available
Amazon MQ	Databases	1	$1.4k	Available
Amplify	Serverless and integration (pay per use)	1	$1.3k	Available
Bedrock AgentCore	AI and ML	1	$1.2k	Available
Transform	Compute (VMs, containers)	1	$495	Not verified
Data Pipeline	Analytics and data platform	1	$327	Not verified

Assumptions

- Recovery target: Recover within 3 hours of a region-wide event. Each service's spend in the chosen accounts is multiplied by its class's second-region share (data/regions.yaml): Compute (VMs, containers) 12%; Databases 45%; Storage (S3, EBS snapshots, files) 80%; Analytics and data platform 25%; Network, security and monitoring 40%; Serverless and integration (pay per use) 5%; Business applications (Connect, WorkSpaces) 15%; AI and ML 5%; Delivery tooling (CI/CD) 5%.
- Replication traffic: 6% of database, storage and analytics spend.
- Second-region prices use an indicative regional price index (EC2 on-demand list prices against us-east-1).
- AWS Support: Enterprise list-price tiers applied to the whole scanned bill plus the new region; enterprise agreements and discounts differ.
- Latency is estimated from distance (fibre ~200 km/ms, routes 1.5x the great circle): measure before committing.
- Service availability: AWS's public regional services table, version 20251113063700 (fetched 2026-10-21), corrected where it is known to be wrong (Amazon Lex available in eu-west-1). Services not listed there are 'not verified' and their effort is raised 25%.
- Region score: parity 55%, latency 20%, price 25%, plus a bonus where the estate already runs.
- People, from today's teams: platform and SRE +10% of 10; security +10% of 3; data +0.5 scaled by √(data spend / US$100k a month), up to 3x; application operations +0.25 per 25 material accounts; DR and resilience +1; at least 1.5 in all. Cost per FTE 165,000 USD a year.
- Recovery times (RTO, RPO) per service: indicative hours by failover type (native, manual, rebuild) and DR strategy (data/regions.yaml), longer for large databases, storage and analytics under backup and restore or rebuild. Prove each with a failover test.
- Build: catalog-style engineer-weeks per service and account × delivery factor 1.26 (CI/CD × IaC × regulated), plus a second-region landing zone (halved where 5%+ of spend already runs there), DR programme, 12% programme management and 25% contingency.
- Duration: the longer of effort ÷ capacity (50% of the platform and data teams + application engineers, 80% productive) and a 20-week critical path, plus 4 weeks of failover rehearsal. Build cost includes running the half-built region meanwhile.
- Figures are ranges (−25% / +50% on cost, −20% / +40% on time), widened to round numbers.